# Midterm 2023 答案附解析版

1. (1) D

(2) C $\lim\limits_{x\to 0}\frac{f(x)}{\sqrt{|x|}}=0$ $\lim\limits_{x\to 0}\frac{f(x)}{x^2}=0$ $\Rightarrow$ $\lim\limits_{x\to 0}f(0)=0$ $f(0)=0$ 推不出来

可导→连续 $\lim\limits_{x\to 0}f(x)=f(0)=0$

$x\to 0^+$ $\lim\limits_{x\to 0}\frac{f(x)}{\sqrt{x}}=\underbrace{\frac{f(x)-f(0)}{x}}_{\text{exist}}\cdot\underbrace{\sqrt{x}}_{0}\Rightarrow 0$ $\lim\limits_{x\to 0}\frac{f(x)}{x^2}=\frac{f(x)-f(0)}{x}\cdot\frac{1}{x}\not\Rightarrow 0$

(3) D 牛顿迭代公式

(4) B $h(x)=f(g(x))$

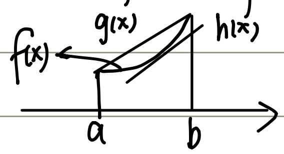
?

$h'(x)=f'(g(x))\,g'(x)\Rightarrow h'(x_0)=0$

$h''(x)=f''(g(x))[g'(x)]^2+f'(g(x))g''(x)=h''(x_0)=\underbrace{f'(a)}_{>0}\underbrace{g''(x_0)}_{<0}<0$

(5) B $g(x)=f(a)+\frac{f(b)-f(a)}{b-a}(x-a)$ $f'(x)\uparrow$

$h(x)=f(c)+f'(c)(x-c)$ $\Downarrow$ $f''(x)>0$ (图象法)




2. (1) 0

(2) $y=2x$, $y=0$(★)

(3) $(\frac{7}{8})^{\frac{4}{3}}$

令 $u=1-x^{\frac{3}{4}}$ $du=-\frac{3}{4}x^{-\frac{1}{4}}dx$ (换元)

$\int_{\frac{7}{8}}^{0}u^{\frac{1}{3}}(-\frac{4}{3})du=-u^{\frac{4}{3}}\Big]_{\frac{7}{8}}^{0}=(\frac{7}{8})^{\frac{4}{3}}$

(4) $2\sqrt{3}+2-\frac{\pi}{3}$

(公式)

$\int_0^{\frac{\pi}{3}}(\sec x+\tan x)^2dx=\int_0^{\frac{\pi}{3}}(\sec^2x+2\sec x\tan x+\tan^2x)dx=\int_0^{\frac{\pi}{3}}2\sec^2x+2\sec x\tan x-1$

$=2\tan x+2\sec x-x\Big]_0^{\frac{\pi}{3}}=2\sqrt{3}+2-\frac{\pi}{3}$

(5) $\frac{1}{16\sqrt{3}}$

$\lim\limits_{x\to 0}\frac{\sqrt{3}-\sqrt{2+\cos x}}{\sin^2 2x}=\lim\limits_{x\to 0}\frac{3-2-\cos x}{2\sqrt{3}\sin^2 2x}=\lim\limits_{x\to 0}\frac{1-\cos x}{2\sqrt{3}\sin^2 2x}=\lim\limits_{x\to 0}\frac{\sin^2 x}{4\sqrt{3}\sin^2 2x}$

△

$=\lim\limits_{x\to 0}\frac{\sin^2 x}{4\sqrt{3}\,4\sin^2x\cos^2x}=\frac{1}{16\sqrt{3}}$

3. $y=\frac{1}{4}x-\frac{5}{4}$

$x^2+2xy^2+3y^4=6$ $y=\frac{1}{4}x-\frac{5}{4}$

$2x+2y^2+2x\cdot 2y\cdot y'+12y^3y'=0$

$y'=\frac{1}{4}$

4. $\int_2^{x^2} \sin(t^3)\,dt + x\sin(x^6)2x$

$x\int_2^{x^2} \sin(t^3)\,dt$ $\sin(t^3)$积不出来

$= \int_2^{x^2} \sin(t^3)\,dt + x\sin(x^6)2x$

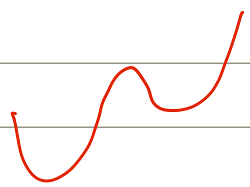
5. $a=b=\frac{1}{2}$

$\lim_{x\to 0^+}\left(\frac{1-\cos x}{x}+a\right)=b$

连续性

$\frac{\sin^2 x}{2x}+a = \frac{\sin x}{x}\cdot\frac{1}{2}\cdot\sin x + a = a$ $\therefore a=b.$

$f'_+(0)=\lim_{x\to 0}\frac{f(x)-f(0)}{x} = \frac{\frac{1-\cos x}{x}+a-b}{x} = \frac{1-\cos x}{x^2} = \frac{\sin^2 x}{2x^2} = \frac{1}{2} = a = b$

6. 

拐点：$x=0,\ 2\pi$

7. $\frac{1}{2}$ (面积相等)

8. 用定义证

$f(x+y)=f(x)f(y)$

$f'(x)=\lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x} = \lim_{\Delta x\to 0}\frac{f(x)f(\Delta x)-f(x)}{\Delta x} = f(x)\lim_{\Delta x\to 0}\frac{f(\Delta x)-1}{\Delta x}$

$=f(x)\lim_{\Delta x\to 0}\frac{1+\Delta x\, g(\Delta x)-1}{\Delta x}$